HAT-P-36b

R-0904

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_HATP-36_240127` · created 2026-07-29T04:04:13+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460336.9923 +/- 0.004 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.129 +/- 0.023
Transit depth (Rp/Rs)^2	1.67 +/- 0.59 [%]
Orbital Inclination (inc)	89.48 +/- 2.23
Airmass coefficient 1 (a1)	1.195 +/- 0.014
Airmass coefficient 2 (a2)	-0.16 +/- 0.11
Scatter in the residuals of the lightcurve fit is	1.18 %
Best Comparison Star	None
Optimal Aperture	4.06
Optimal Annulus	14.85
Transit Duration (day)	0.0971 +/- 0.0027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.129 ± 0.023 vs 0.126 (deviation 0.09σ)
Duration fit vs published	0.0971 d vs 0.0925 d (deviation 1.19σ)
Mid-time O–C	-2.1 min
Depth SNR	3.9
Residual scatter	1.18 %

Light curve

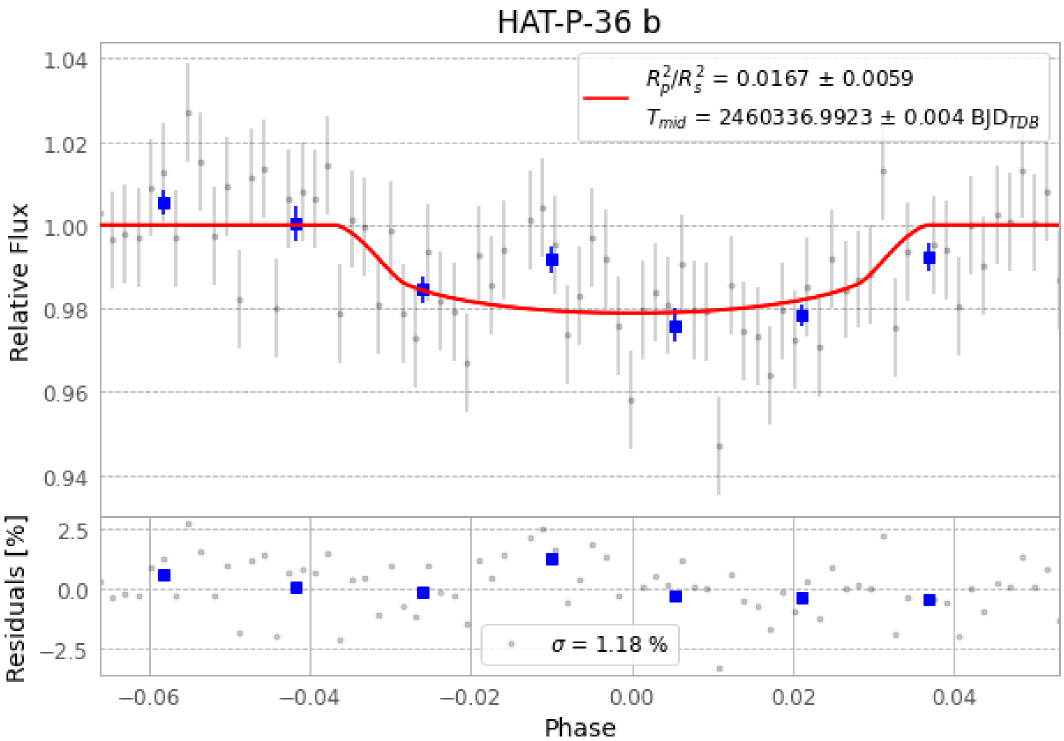

Figure 1 — FinalLightCurve_HAT-P-36 b_2024-01-27.png (SHA-256 cba5541647bb2b0a…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [382, 262], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 c46562289f60bc5e…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9`

Data provenance

77 FITS frames (51 MB), `HATP-36240127093616.FITS` → `HATP-36240127132415.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-240127.log (2bdc18deaab8…), FinalLightCurve_HAT-P-36 b_2024-01-27.png (cba5541647bb…), FinalLightCurve_HAT-P-36 b_2024-01-27.csv (0b2923e51cbc…)

Compute resources

Wall time 125.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)